

CSCE 46103/56103 Artificial Intelligence – Fall 2026

Exact Inference in Bayesian Networks

Jungtaek Kim

`jungtaek.kim@uark.edu`

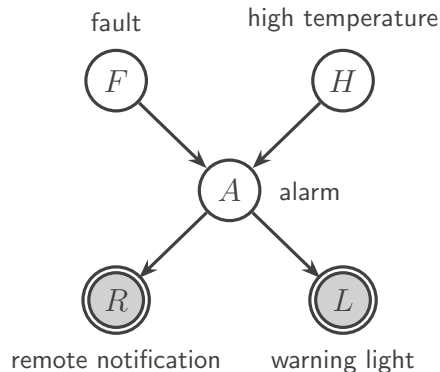
University of Arkansas

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What Is Inference?

- ▶ Inference uses what we observe and what we already know to reason about what we do not know.
- ▶ In a Bayesian network, this means using the model and the available evidence to compute how likely different possibilities are.
- ▶ This matters because observations are often incomplete or noisy. These probabilities help us diagnose faults, predict outcomes, and make decisions under uncertainty.

Inferring a Fault from Sensor Readings



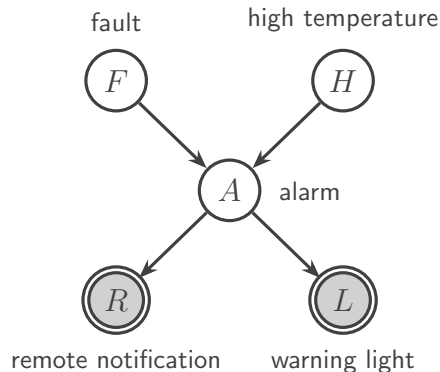
The Previous Question

The notification and warning light are on.
How likely is a fault?

$$P(F \mid r, l)$$

Temperature and alarm status are unobserved.

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Goal

Compute the posterior exactly, then reuse intermediate results to avoid repeated work.

Shading and a double border mark observed variables.

Query, Evidence, and Hidden Variables

Role	General notation	Sensor query
Query variable	X	F
Observed variables	$\mathbf{E} = \mathbf{e}$	$R = \text{true}, L = \text{true}$
Hidden variables	$\mathbf{Y}$	H, A

- Keep every possible value of the query variable.
- Fix observed variables to their evidence values.
- Sum over all values of the hidden variables.

Notation

For the Boolean sensor variables, uppercase denotes a variable and lowercase its true value: f means $F = \text{true}$ and $\neg f$ means $F = \text{false}$.

The Local Probability Tables

Roots and Alarm

$$P(f) = 0.01, \quad P(h) = 0.10.$$

F	H	$P(a \mid F, H)$
true	true	0.99
true	false	0.89
false	true	0.41
false	false	0.01

Notification and Light

A	$P(r \mid A)$	$P(l \mid A)$
true	0.80	0.90
false	0.10	0.05

The roots F, H are independent. Each omitted Boolean probability is the complement.

$$P(F, H, A, R, L) = P(F)P(H)P(A \mid F, H)P(R \mid A)P(L \mid A).$$

Inference by Enumeration

Sum, Then Normalize

$$q(x) = \sum_{\mathbf{y}} P(x, \mathbf{e}, \mathbf{y}), \quad P(x \mid \mathbf{e}) = \frac{q(x)}{\sum_{x'} q(x')}.$$

The denominator is $P(\mathbf{e})$; the posterior requires $P(\mathbf{e}) > 0$.

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For each $x \in \{f, \neg f\}$ in our sensor model,

$$q(x) = \sum_{h' \in \{h, \neg h\}} \sum_{a' \in \{a, \neg a\}} P(x)P(h')P(a' \mid x, h')P(r \mid a')P(l \mid a').$$

- Four hidden assignments for each value of F .
- Each term is a product of five local probabilities.

Four Hidden Assignments

Each row fixes H , A and includes the prior probability of the indicated value of F .

H	A	$P(f, H, A, r, l)$	$P(\neg f, H, A, r, l)$
h	a	0.00071280	0.02922480
h	$\neg a$	0.00000005	0.00029205
$\neg h$	a	0.00576720	0.00641520
$\neg h$	$\neg a$	0.00000495	0.00441045
Sum		0.00648500	0.04034250

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One Row

$$P(f, \neg h, a, r, l) = 0.01(0.90)(0.89)(0.80)(0.90) = 0.00576720.$$

Normalization and the Posterior

$$q(f) = 0.006485, \quad q(\neg f) = 0.0403425.$$

$$P(r, l) = q(f) + q(\neg f) = 0.0468275,$$

$$P(f \mid r, l) = \frac{0.006485}{0.0468275} \approx 0.138487,$$

$$P(\neg f \mid r, l) = 1 - P(f \mid r, l) \approx 0.861513.$$

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From Prior to Posterior

The fault probability rises from 1% to about 13.85%. Observing both signals does not make a fault certain.

Exact inference computes the posterior specified by the model; displayed decimals are rounded.

Enumeration and Repeated Work

The same calculation can be written as nested sums:

$$q(x) = P(x) \sum_{h'} P(h') \sum_{a'} P(a' \mid x, h') \underbrace{P(r \mid a') P(l \mid a')}_{\text{depends only on } a'}.$$

- ▶ In topological order, multiply the local probability at each variable.
- ▶ Use the assigned value for query/evidence variables; branch and sum for hidden variables.
- ▶ Repeated branches can recompute the same subexpressions.

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Reuse Through Distributivity

Store a result once for the variables it depends on. Variable elimination reorganizes the same sums and products without changing the answer.

Factors, Scopes, and Evidence

A factor is a table of values indexed by a set of variables: its scope.

Original factor	After observing r, l	Remaining scope
$P(F)$	$P(F)$	$\{F\}$
$P(H)$	$P(H)$	$\{H\}$
$P(A \mid F, H)$	$P(A \mid F, H)$	$\{F, H, A\}$
$P(R \mid A)$	$P(r \mid A)$	$\{A\}$
$P(L \mid A)$	$P(l \mid A)$	$\{A\}$

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Restriction, Not Summation

Evidence selects the observed entries. The resulting factor need not sum to one over its remaining variables.

Factor Multiplication

Multiply entries that agree on shared variables. The product's scope is the union of the input scopes.

$$\phi(F, H, A) = P(H)P(A \mid F, H).$$

First scope	Second scope	Product scope
$\{H\}$	$\{F, H, A\}$	$\{F, H, A\}$

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One Product Entry

$$\phi(f, h, a) = P(h)P(a \mid f, h) = 0.10(0.99) = 0.099.$$

For Boolean F, H, A , this product has $2^3 = 8$ entries. Factor multiplication is not matrix multiplication.

Summing Out a Variable

To remove H from $\phi(F, H, A)$, sum over its values while holding F, A fixed:

$$m(F, A) = \sum_{h'} \phi(F, h', A).$$

One Result Entry

$$\begin{aligned} m(f, a) &= \phi(f, h, a) + \phi(f, \neg h, a) \\ &= 0.10(0.99) + 0.90(0.89) = 0.90. \end{aligned}$$

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- ▶ The resulting scope is $\{F, A\}$: four Boolean entries.
- ▶ Multiply all current factors containing a variable before summing it out.
- ▶ Factors that do not contain that variable remain separate.

The Variable Elimination Procedure

1. Create all CPT factors and restrict them to the evidence.
2. Choose an order for the hidden variables.
3. For each hidden variable Z , collect all current factors containing Z , multiply them, and sum out Z :

$$g(\mathbf{U}) = \sum_z \prod_{\phi: Z \in \text{scope}(\phi)} \phi.$$

Replace the collected factors with g ; $\mathbf{U}$ is their remaining scope.

4. Multiply the remaining factors over the query variable.
5. Normalize the result.

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Preserving the Query

Eliminate hidden variables only. Evidence values are fixed, and the query variable remains in the final result.

The Two Observed Signals

For our query, the restricted evidence factors share the scope $\{A\}$:

$$\ell(A) = P(r \mid A)P(l \mid A).$$

A	$P(r \mid A)$	$P(l \mid A)$	$\ell(A)$
a	0.80	0.90	0.720
$\neg a$	0.10	0.05	0.005

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A Likelihood Factor

R, L are conditionally independent given A , so $\ell(A) = P(r, l \mid A)$. This is not a posterior distribution over A .

The current factors are $P(F)$, $P(H)$, $P(A \mid F, H)$, and $\ell(A)$.

First Elimination: Temperature

Choose the hidden-variable order H, A . Only $P(H)$ and $P(A \mid F, H)$ contain H :

$$m(F, A) = \sum_{h'} P(h') P(A \mid F, h').$$

F	$m(F, a)$	$m(F, \neg a)$
f	0.90	0.10
$\neg f$	0.05	0.95

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$\neg f$	0.05	0.95

The Remaining Factors

$$P(F), \quad m(F, A), \quad \ell(A).$$

Here $m(F, A) = P(A \mid F)$, recovering the alarm rates from the earlier probability lecture.

Second Elimination: Alarm

Multiply the two factors containing A , then sum out A :

$$s(F) = \sum_{a'} m(F, a') \ell(a').$$

F	$m(F, a)\ell(a)$	$m(F, \neg a)\ell(\neg a)$	$s(F)$
f	$0.90(0.72) = 0.648$	$0.10(0.005) = 0.0005$	0.64850
$\neg f$	$0.05(0.72) = 0.036$	$0.95(0.005) = 0.00475$	0.04075

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$\neg f$	$0.05(0.72) = 0.036$	$0.95(0.005) = 0.00475$	0.04075

The Remaining Factors

$$P(F), \quad s(F).$$

The hidden variables are gone. The likelihood $s(F) = P(r, l \mid F)$ must still be multiplied by the prior.

The Same Posterior, Reused Computation

$$q(f) = P(f)s(f) = 0.01(0.6485) = 0.006485,$$
$$q(\neg f) = P(\neg f)s(\neg f) = 0.99(0.04075) = 0.0403425.$$

$$P(f \mid r, l) = \frac{0.006485}{0.006485 + 0.0403425} \approx 13.85\%.$$

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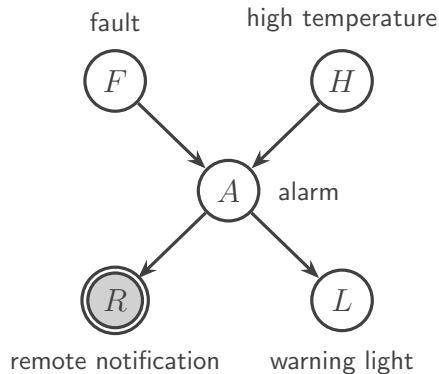
$$P(f \mid r, l) = \frac{0.006485}{0.006485 + 0.0403425} \approx 13.85\%.$$

Why the Answer Is Unchanged

$$\sum_{h', a'} P(h')P(a' \mid F, h')\ell(a') = \sum_{a'} \ell(a') \underbrace{\sum_{h'} P(h')P(a' \mid F, h')}_{m(F, a')}.$$

Variable elimination rearranges finite sums and products and stores intermediate factors. It makes no sampling approximation.

Irrelevant Variables and Pruning



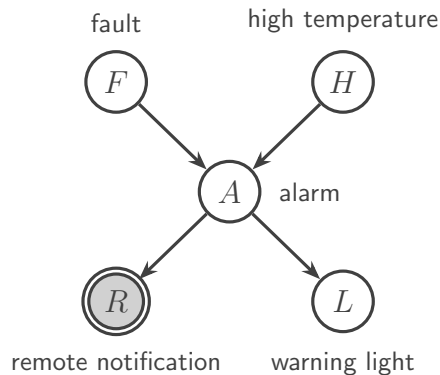
For the query $P(F \mid r)$, the warning light is an unobserved leaf:

$$\sum_L P(L \mid A) = 1.$$

Its factor disappears when L is summed out.

Prune unobserved leaves outside the query and evidence; repeat as new leaves appear.

Irrelevant Variables and Pruning



For the query $P(F \mid r)$, the warning light is an unobserved leaf:

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Its factor disappears when L is summed out.

Prune unobserved leaves outside the query and evidence; repeat as new leaves appear.

$$P(F \mid r) \propto P(F) \sum_{H,A} P(H) P(A \mid F, H) P(r \mid A).$$

Elimination Order and Factor Size

For $P(L \mid r)$, eliminate F, H, A while retaining L .

Order F, H, A

$$g(H, A) = \sum_F [P(F) \cdot P(A \mid F, H)].$$

First step	Entries
Product on $\{F, H, A\}$	8
Result on $\{H, A\}$	4

Order A, F, H

$$g(F, H, L) = \sum_A [P(A \mid F, H) \cdot P(r \mid A)P(L \mid A)].$$

First step	Entries
Product on $\{F, H, A, L\}$	16
Result on $\{F, H, L\}$	8

All variables are Boolean. Both orders give the same normalized answer.

Factor Size and Induced Width

Connect variables that share a factor. Eliminating a variable connects all its remaining neighbors.

Width of an Elimination Order

For a single-variable query kept last, let w be the largest number of remaining neighbors of any eliminated variable.

$$\underbrace{d^{w+1}}_{\text{largest product table}} \longrightarrow \underbrace{d^w}_{\text{result after summing out}},$$

where each variable has at most d values; these are worst-case bounds.

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where each variable has at most d values; these are worst-case bounds.

- ▶ Treewidth is the smallest induced width over all elimination orders.
- ▶ The largest factors control the cost: even Boolean tables grow exponentially with width.
- ▶ A practical heuristic chooses the hidden variable producing the smallest next factor.

Structure and Tractability

Polytrees

There is at most one undirected path between any two nodes.

With suitable elimination or message passing, exact marginal inference is linear in total tabular CPT size.

Our sensor network is a polytree. An arbitrary order can still create unnecessary factors.

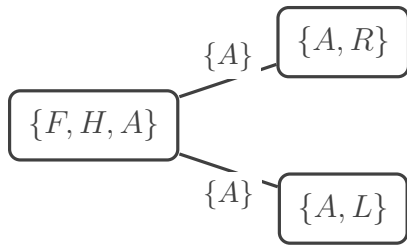
General Networks

Multiple paths can force large intermediate factors.

Exact inference is $\#P$ -hard in general; variable elimination does not make every network efficient.

Low treewidth and small variable domains can still make a large network manageable.

Clustering for Repeated Queries



A junction tree for the sensor network. Edge labels are shared variables.

Clusters containing a variable must form a connected subtree. Large clusters still make exact inference expensive.

1. Assign CPTs to clusters containing their variables; incorporate evidence.
2. Pass messages so neighbors agree on shared-variable marginals.
3. Reuse the results for several marginal queries with the same evidence.

Checkpoint: A Notification Without a Light

Compute $P(F \mid r, \neg l)$ using the same sensor network.

After Eliminating H

$P(f) = 0.01.$	
F	$P(a \mid F)$
true	0.90
false	0.05

Sensor CPTs

A	$P(r \mid A)$	$P(l \mid A)$
true	0.80	0.90
false	0.10	0.05

The model gives $R \perp\!\!\!\perp L \mid A$.

1. Form the likelihood factor $P(r \mid A)P(\neg l \mid A)$.
2. Sum out A and include $P(F)$ to obtain both unnormalized weights.
3. Normalize. Does the fault probability rise or fall from its 1% prior?

Checkpoint Solution

The evidence likelihood factor is

$$P(r, \neg l \mid a) = 0.80(0.10) = 0.08, \quad P(r, \neg l \mid \neg a) = 0.10(0.95) = 0.095.$$

The unnormalized weights are

$$\begin{aligned} q(f) &= 0.01 [0.90(0.08) + 0.10(0.095)] = 0.000815, \\ q(\neg f) &= 0.99 [0.05(0.08) + 0.95(0.095)] = 0.0933075. \end{aligned}$$

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The unnormalized weights are

$$\begin{aligned} q(f) &= 0.01 [0.90(0.08) + 0.10(0.095)] = 0.000815, \\ q(\neg f) &= 0.99 [0.05(0.08) + 0.95(0.095)] = 0.0933075. \end{aligned}$$

$$P(f \mid r, \neg l) = \frac{0.000815}{0.000815 + 0.0933075} \approx 0.00866 = 0.866\%.$$

This evidence is slightly more likely without an alarm, so the fault probability falls below its prior.

Exact Inference: Main Takeaways

- ▶ State the query, fix the evidence, and sum out hidden variables.
- ▶ Enumeration sums complete assignments; variable elimination reuses intermediate factors.
- ▶ Elimination order changes factor sizes and computation, while preserving the answer.
- ▶ Graph structure determines whether exact inference is practical.

Thank You

Any questions?

References

S. J. Russell and P. Norvig. *Artificial Intelligence: A Modern Approach*. Pearson, 4th edition, 2020. URL <https://aima.cs.berkeley.edu/>.